

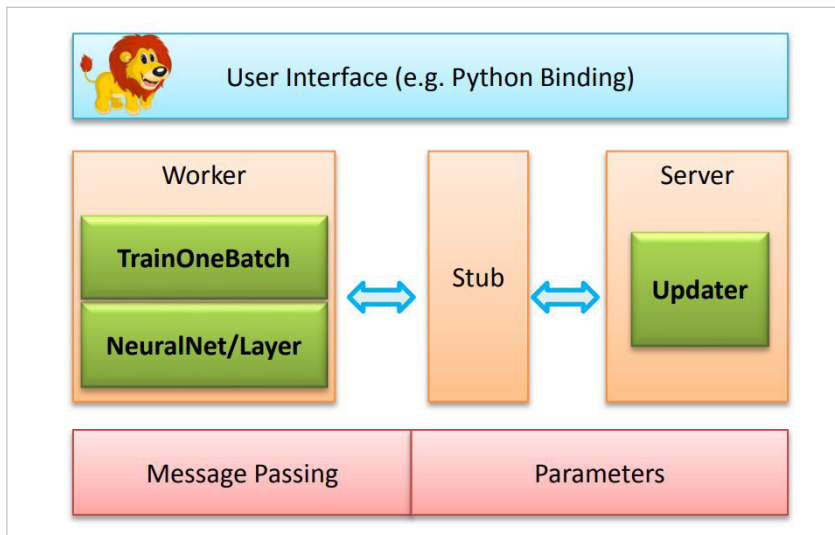


Figure 1: SINGA

support and integrates with cluster management software.

SINGA applies stochastic gradient descent (SGD) algorithm to update parameters of deep learning models. The workload is dispersed over worker and server units. At each round, all workers call TrainOneBatch function to compute parameter gradients. TrainOneBatch pulls a NeuralNet object on behalf of the neural net, and calls the layers of the NeuralNet in a definite order. The resulting gradients are sent to the local stub that collects the requests and forwards them to equivalent servers for updating. Servers respond to workers with the improved parameters for the next iteration. To start a training job, users submit a job configuration consisting of four components:

- 1) A NeuralNet describing the neural net structure with detailed layer settings and their connections.
- 2) A TrainOneBatch algorithm that is custom-made for different model categories.
- 3) An Updater defining the protocol for updating parameters at the server side.
- 4) A Cluster topology specifying the distributed architecture of workers and servers.

Features

This open source platform is designed to scale up deep learning to thousands of GPUs, which means you can process massive amounts of data with improved accuracy. Developers can tailor Apache SINGA to their specific needs, with customisable features and modular design. Here are some of SINGA's key features.

User-friendly: It comes with an intuitive programming model based on the layer abstraction, and many built-in layers are already provided. It is easy for users to implement their deep learning algorithms without much knowledge of the underlying distributed platform. Different types of popular deep learning models can be expressed and trained using SINGA.

Scalable: It provides a general architecture to exploit the scalability of different training frameworks. Synchronous training frameworks improve the efficiency of training iterations while asynchronous training frameworks improve the convergence rate. SINGA also offers diverse

partitioning schemes to parallelise the training of large models, feature dimension or hybrid partitioning.

Other features: It supports training on a single node (i.e., process) with multiple GPU cards. It is integrated with the Mesos framework using Docker, which bundles Mesos and SINGA.

Installation guidelines using Docker

Assuming Docker is already installed on the system, add your user to the Docker group to run Docker commands without sudo.

CPU-only

```
$ docker run -it apache/singa:X.Y.Z-cpu-ubuntu16.04 /bin/bash
```

With GPU enabled, install Nvidia-Docker after installing Docker:

```
$ nvidia-docker run -it apache/singa:X.Y.Z-cuda9.0-cudnn7.4.2-ubuntu16.04 /bin/bash
```

Run the command given below:

```
version-{cpu gpu}[-devel]
```

Explanation

Version: SINGA version

cpu: The image cannot run on GPUs

gpu: The image can run on Nvidia GPUs

devel: Indicator for development

OS: OS version number

Apache SINGA opens up a new realm of possibilities in deep learning. Its documentation and code are available on the Apache Incubator website. A quick start guide and some sample examples are also provided. 

Reference

- <https://svn.apache.org/>

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